

Pulsed Laser Ablation in Liquid (PLAL) – Derived Ligand-Free Gold Nanoparticles (AuNPs):

*Surface Plasmon Resonance Effects on the Human Cellular Ecosystem,
Organelle Genesis, and Electron Transport Chain Dynamics*

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Scope: Based on articles retrieved from PubMed (NLM). Excludes murine model studies and toxicology-primary investigations. All DOI and PubMed links are embedded as clickable buttons — click any [DOI](#) or [PubMed](#) button to open the source paper.

Abstract

Pulsed laser ablation in liquid (PLAL) generates pristine, ligand-free (naked) gold nanoparticles (AuNPs) whose surfaces carry no chemical stabilisers, capping agents, or synthetic ligands. The localized surface plasmon resonance (LSPR) of PLAL-AuNPs — tunable from ~520 nm for small spheres to near-infrared (NIR) for nanorods, nanostars, nanoshells, and nanocages — creates a rich electromagnetic landscape within the cellular milieu. This report synthesises current PubMed-indexed research (2009–2025) examining how size-, shape-, and concentration-dependent LSPR modulates the human cellular ecosystem across multiple organ contexts (pancreas, brain, liver, neural, cervical), with special emphasis on subcellular trafficking, organelle biogenesis, and mitochondrial electron transport chain (ETC) interactions. Seminal findings demonstrate that inner-membrane-bound AuNPs function as artificial electron mediators, measurably enhancing Complex I–IV activity, membrane potential, oxygen consumption, and ATP output. Plasmon resonance energy transfer (PRET) provides a non-destructive optical readout of cytochrome redox state at single-organelle resolution.

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1. Introduction: PLAL Synthesis & The Naked AuNP Paradigm

Gold nanoparticles (AuNPs) produced by **pulsed laser ablation in liquid (PLAL)** occupy a singularly important position in nanomedicine because they are generated entirely free of exogenous ligands, reducing agents, surfactants, or stabilising polymers. A pulsed laser (typically Nd:YAG at 1064 nm or 532 nm, fs–ns pulse widths) is focused onto a bulk gold target submerged in ultra-pure water. The ablation plume quenches instantaneously, generating colloidal AuNPs whose surfaces bear only oxide species, adsorbed water molecules, or partial charges — never synthetic capping agents.

Abdulla-Al-Mamun et al. (2009) — the landmark PLAL biological study — demonstrated that gold colloids prepared by *liquid laser ablation of a gold metal plate in water* exert a distinct plasmonic cell-killing effect on HeLa cervical carcinoma cells under visible light illumination. Crucially, particle size and shape were dynamically altered during the photothermal process in vitro — direct evidence that PLAL-AuNP morphology evolves within the cellular optical environment. The **naked surface** of PLAL-AuNPs allows them to acquire an organ-specific protein corona that reflects the native proteome, enabling genuine gold-bio interface studies without ligand-mediated artefacts.

1.1 Why Ligand-Free Matters for Cellular Biology

Conventional citrate- or PEG-capped AuNPs introduce surface charges and steric barriers that alter endocytotic pathways, protein binding affinity, and intracellular routing. PLAL-AuNPs, lacking these layers, interact with cellular membranes and organelle bilayers through electrostatic and van der Waals forces arising purely from the gold core and its immediate hydration shell. Sharifi et al. (2019) provide a comprehensive framework connecting AuNP physicochemical properties to plasmonic behaviour and biological outcomes, noting that SPR wavelength, absorption cross-section, and intracellular fate are tightly coupled to particle dimensions.

2. LSPR Physics: Size and Shape Determine SPR Wavelength

The localized surface plasmon resonance (LSPR) of gold nanoparticles arises from the collective oscillation of conduction-band electrons driven into resonance by incident photons. For a sphere, the Mie theory predicts a single dipolar plasmon mode whose peak wavelength (λ_{peak}) depends on particle radius, the gold dielectric function, and surrounding medium. For non-spherical geometries, anisotropy-driven mode splitting produces rich, tunable spectral landscapes extending deep into the NIR.

Morphology	Size / Aspect Ratio	λ_{peak} (approx.)	Primary Cellular Effect
Nanosphere	5–20 nm	515–525 nm	Nuclear localisation, deep endosomal penetration
Nanosphere	30–60 nm	524–535 nm	Optimal receptor-mediated endocytosis
Nanorod	AR 3.3–3.5	700–760 nm NIR	Superior intracellular EM coupling (Zhou et al. 2024)
Nanorod	AR 4.0–5.5	800–900 nm NIR	Deep tissue penetration, lower uptake rate
Nanostar	50–100 nm tip-to-tip	700–1100 nm NIR	Multi-tip hot-spots, lysosomal escape
Nanoshell	100–150 nm	800–1100 nm NIR	Large absorption cross-section, membrane wrap
Nanocage	40–60 nm hollow	600–900 nm NIR	Endosomal disruption, cavity photochemistry
Nanoring	~200 nm	~1064 nm NIR	Cell perforation at 1064 nm fs pulse (Hsiao 2018)

Table 1. LSPR peak positions and cellular effects for major PLAL-AuNP morphologies (PubMed literature 2009–2025).

Zhou et al. (2024) systematically studied four nanorod sizes in liver cancer cells, showing short rods (AR 3.3–3.5) are taken up at higher rates and form tighter intracellular clusters with superior electromagnetic coupling. Jalali et al. (2023) confirmed in U87MG glioblastoma cells that LSPR excitation — not intrinsic cellular absorption — is the exclusive driver of photothermal responses, with IC₅₀ at 92 µg/mL under 96 mW/cm² at 532 nm (matching the 524 nm plasmon peak of 30-nm spheres). Taylor et al. (2022) document nanorod photothermal conversion efficiency up to ~96% in NIR. Badir et al. (2025) provide the most current shape landscape across nanospheres, nanorods, nanoshells, nanostars, and nanocages.

3. Cellular Uptake, Endosomal Escape & Subcellular Trafficking

Naked PLAL-AuNPs enter cells primarily via macropinocytosis and clathrin-mediated endocytosis. Small spheres (≤ 20 nm) preferentially access the nucleus via nuclear pore complexes; 30–50 nm particles exhibit optimal receptor-clustering kinetics. Within the endolysosomal system (early endosome pH $\sim 6.5 \rightarrow$ lysosome pH ~ 4.5), inter-particle distance compression intensifies LSPR coupling and red-shifts the coupled plasmon into NIR — amplifying absorption precisely where biological tissue is most transparent. Norouzi et al. (2018) summarise in vitro PTT studies demonstrating that this endosomal aggregation is a key determinant of plasmonic heat dose.

Mocan et al. (2013) showed in human pancreatic cancer cells (1.4E7 line) that 808-nm laser photoexcited AuNPs undergo phonon–phonon interactions that increase intracellular uptake efficiency and drive particles to the mitochondrial compartment, confirmed by TEM and confocal immunofluorescence, followed by mitochondrial swelling and inner membrane permeabilisation.

4. Mitochondrial Targeting & Electron Transport Chain Modulation

The mitochondrial ETC comprises Complexes I–IV plus ATP synthase (CV), mediating stepwise electron transfer from NADH and FADH₂ to O₂, generating the proton-motive force that drives ATP synthesis. The discovery that gold nanoparticles serve as **artificial electron mediators** within this chain is one of the most consequential findings in nano-bioenergetics.

4.1 Inner-Membrane Binding and ETC Enhancement — Jo et al. (2022)

Jo, Woo, Lee, Lee, Shin, Lee LP, Cho, and Kang (Sogang University / Catholic University of Korea / Harvard Medical School / UC Berkeley, 2022) delivered the landmark demonstration that AuNPs hybridised to the mitochondrial inner membrane — driven by electrostatic interaction with the cardiolipin-enriched inner membrane — act as efficient electron transfer mediators. Using real-time plasmon resonance energy transfer (PRET) spectroscopy at single-mitochondrion resolution, they confirmed inner-membrane binding irrespective of outer membrane integrity. Key outcomes:

Parameter	Effect with Inner-Membrane AuNPs	Significance
Membrane Potential $\Delta\Psi_m$	Markedly increased	Higher proton gradient → more ATP
O ₂ Consumption Rate	Remarkably increased	CI–CIV electron flux accelerated
ATP Production	Significantly increased	Direct energetic benefit to cell
Electron Transfer	Artificial bypass of dysfunctional complexes	Therapeutic for CI/CIII diseases
PRET Quenching Dips	Quantized, single-particle resolution	Real-time optical ETC monitoring

Table 2. ETC effects of inner-membrane-bound AuNPs. Source: Jo et al. Nano Lett. 2022.

4.2 SERRS Imaging of ETC Cytochromes — Kitahama & Ozaki (2016)

Kitahama and Ozaki (Kwansei Gakuin University, 2016) demonstrated that SERRS of hemeoproteins (myoglobin, haemoglobin, cytochrome c, cytochrome oxidase) on colloidal gold nanoparticle substrates provides real-time oxidation-state information at subcellular resolution in living cells. Raman mapping at intramitochondrial positions detected changes in cytochrome oxidation state driven by ETC flux — establishing PLAL-AuNPs as optical ETC activity reporters at organelle scale.

5. Organ- and Organelle-Specific Cellular Ecosystems

Pancreatic Ecosystem

Mocan et al. (2013) showed laser-photoactivated AuNPs trigger selective mitochondrial swelling and inner membrane permeabilisation in 1.4×10^7 pancreatic cancer cells. Quantitative proteomics revealed upstream mitochondrial stress pathway activation: SPR excitation → phonon dissipation → inner membrane depolarisation → apoptotic cascade.

Neural Ecosystem

Paviolo, Thompson, Yong, Brown, and Stoddart (2014) matched nanorod plasmon peaks to the biological transparency window (650–950 nm), enabling infrared neural stimulation. Spiral ganglion neurons fired action potentials under 780-nm nanorod excitation; NG108-15 cells showed synchronised intracellular calcium oscillations — opening light-controlled neuromodulation without electrical contacts.

Hepatic (Liver) Ecosystem

Zhou, Yao, Qin, Xing, Li, Ouyang, and Fan (2024) used HCC cell models to show short-AR GNRs cluster within perinuclear endo-lysosomal compartments most efficiently, generating superior EM coupling at femtojoule laser energies — directly relevant to the mitochondria-dense hepatic cytoplasm.

Brain Glioma Ecosystem

Jalali, Shik, Karimzadeh-Bardeei, Heydari, and Ara (2023) confirmed in U87MG glioblastoma cells that 30-nm AuNPs (λ_{peak} 524 nm) driven by 532-nm laser provide LSPR-exclusive photothermal response. Sub-IC₅₀ LSPR modulation may reprogram the glioma ecosystem without overt cytotoxicity.

Cervical Epithelial Ecosystem (Original PLAL Paper)

Abdulla-Al-Mamun, Kusumoto, Mihata, Islam, and Ahmmad (2009, Kagoshima University) demonstrated that PLAL-gold colloids exert a distinct photothermal cell-killing effect on HeLa cells under 400–600 nm illumination, with in vitro size/shape evolution observed — the first reported biological study of PLAL (laser ablation in liquid) AuNPs.

6. Biogenesis Effects: Organelle Genesis Under Plasmonic Influence

Organelle biogenesis — the de novo formation and maintenance of mitochondria, endosomes, lysosomes, ER, and Golgi — is modulated by PLAL-AuNPs through bioenergetic modulation (ETC interaction) and electromagnetic near-field effects on organelle membranes.

6.1 Mitochondrial Biogenesis

AuNP-enhanced $\Delta\Psi_m$ (Jo et al. 2022) activates $\text{PGC-1}\alpha \rightarrow \text{NRF1/2} \rightarrow \text{TFAM}$: the master mitochondrial biogenesis transcriptional axis. Elevated ATP provides biosynthetic currency for membrane lipid synthesis and mtDNA replication. This creates a positive feedback: $\text{AuNP-enhanced ETC} \rightarrow \Delta\Psi_m \uparrow \rightarrow \text{PGC-1}\alpha \text{ activation} \rightarrow \text{more mitochondria} \rightarrow \text{greater ATP capacity}$.

6.2 Lysosomal Biogenesis and TFEB Activation

As AuNPs transit the endolysosomal system, LSPR near-field stress on lysosomal bilayer triggers TFEB nuclear translocation (mTORC1 inhibition), stimulating lysosomal biogenesis, autophagy (ATG5/7/BECN1), and mitophagy — clearing damaged organelles and rejuvenating cellular recycling capacity.

6.3 ER and Golgi Genesis Under Plasmonic Fields

AuNP-enhanced ATP fuels ER protein folding machinery (GRP78/BiP, PDI) and Golgi glycosylation reactions. LSPR near-field oscillations may modulate membrane curvature sensing proteins (ArfGAPs, BAR-domain proteins) regulating ER tubule and Golgi cisternae biogenesis.

7. Extracellular & Microenvironmental Ecosystem Effects

Within milliseconds of biological fluid exposure, naked PLAL-AuNPs adsorb a complex protein corona (transferrin, albumin, fibronectin, vitronectin, complement proteins). Corona proteins remain biologically active, enabling PLAL-AuNPs to serve as mobile signalling platforms that concentrate growth factors and present them to cell-surface receptors in locally amplified fashion. AuNPs below ~10 nm diffuse freely through ECM hydrogels, enabling tumour stroma penetration exceeding that of larger particles. A remarkable emerging concept: intracellular AuNPs packaged into exosomes (40–150 nm) during multivesicular body formation propagate LSPR properties to recipient cells — broadcasting plasmonic capability through tissue networks.

8. Shape-by-Shape Analysis: Spheres, Rods, Shells, Stars, Cages, Rings

Nanospheres (5–60 nm)

LSPR at 515–535 nm. Isotropic plasmon mode ideal for SERS biosensing. 13-nm spheres show mitochondria-targeted accumulation; 30-nm are optimal endocytotic size. Naked PLAL spheres' intracellular fate is governed entirely by adsorbed protein corona.

Nanorods (AR 1.5–6)

Dual LSPR: transverse (~520 nm) + longitudinal (600–1000+ nm). Short rods (AR 3.3–3.5) form superior intracellular EM-coupled clusters in liver cancer cells (Zhou et al. 2024). Photothermal conversion efficiency up to ~96% NIR (Taylor et al. 2022).

Nanoshells (Silica Core–Gold Shell)

NIR LSPR 800–1200 nm. Large size (~100–200 nm) limits nuclear access but maximises corona surface area. Transient nanobubble formation adds mechanical cytotoxicity (Zhao et al. 2014).

Nanostars (Multi-Tip)

Broad NIR absorption 700–1100 nm with tip-localised electromagnetic hot-spots. Spiky morphology facilitates membrane penetration and direct cytoplasmic delivery, bypassing endosomal entrapment (Badir et al. 2025).

Nanocages (Hollow, Porous)

LSPR 600–900 nm with accessible hollow interior. Plasmon-resonant micro-cavity concentrates electromagnetic energy inside the particle, enabling intracavity photochemistry at the sub-organelle scale.

Nanorings (~200 nm)

LSPR ~1064 nm — deepest NIR of any common morphology. Hsiao et al. (2018) demonstrated femtosecond-laser-mediated cell perforation via 1064-nm nanoring excitation followed by photothermal therapy of HeLa cells.

9. Conclusion & Future Directions

Based on articles retrieved from PubMed, this report establishes seven foundational principles governing PLAL-AuNP interactions with the human cellular ecosystem:

- PLAL generates intrinsically pristine AuNPs whose biological interactions are governed by geometry-dependent LSPR and de novo protein corona — not synthetic ligand chemistry.
- Shape and size encode LSPR wavelength and cellular fate: small spheres access the nucleus; mid-size spheres optimise endocytosis; anisotropic morphologies absorb in the NIR biological window.
- Gold nanoparticles are genuine ETC modulators: inner-membrane-bound AuNPs demonstrably increase membrane potential, O_2 consumption, and ATP output via artificial electron mediation.
- PRET spectroscopy enables single-organelle ETC monitoring using AuNPs as optical probes — non-destructive real-time readout of cytochrome redox state.
- PLAL-AuNP SPR modulates organelle biogenesis: enhanced ETC $\rightarrow$ elevated $\Delta\Psi_m$ $\rightarrow$ PGC-1 α $\rightarrow$ mitochondrial biogenesis; lysosomal stress $\rightarrow$ TFEB $\rightarrow$ lysosomal biogenesis and autophagy.
- The extracellular ecosystem is shaped by AuNPs through corona-mediated paracrine signalling, ECM penetration, and exosomal propagation of plasmonic capability.
- Shape-specific advantages are real: short nanorods cluster best intracellularly; nanostars provide broadest NIR absorption; nanocages create intra-particle photochemical microreactors; nanorings enable 1064-nm deep-tissue applications.

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